Isotopes and Radioisotopes

Every atom of carbon has six protons. That is what makes it carbon. But not every carbon atom is identical. Some carbon atoms have six neutrons. Some have seven. Some have eight. These different versions of the same element are called isotopes. Some isotopes are stable and last forever. Others are unstable and decay, releasing radiation. These radioactive isotopes – called radioisotopes – can be both beneficial and harmful. They can destroy cancer cells, but they can also cause cancer. They can generate electricity in nuclear power plants, but they can also contaminate the environment for thousands of years. Yet most people never learn how these powerful substances work.

Chemistry is not merely a subject taught in classrooms. It is the science of matter and its transformations. Isotopes explain why some elements are radioactive, how scientists date ancient artifacts, and how doctors diagnose and treat diseases. Understanding isotopes means understanding the hidden power inside the nucleus of atoms – power that can save lives or cause harm depending on how it is used.

This reading material explores what isotopes are, how they differ from one another, the properties of natural and human-made radioisotopes (including their types of radiation and half-lives), and the important uses of radioisotopes in medicine.

Part 1: What Are Isotopes and Why Do They Matter?

Definition of Isotopes

Isotopes are atoms of the same element that have the same number of protons but a different number of neutrons. Because they have the same number of protons, they are the same element. Because they have different numbers of neutrons, they have different mass numbers.

The Central Idea: Same Element, Different Mass

All isotopes of a given element behave almost identically in chemical reactions because chemical behavior depends on electrons (specifically, the number and arrangement of electrons). However, isotopes differ in their nuclear properties – some are stable, some are radioactive.

Why Isotopes Matter

Aspect	Explanation
Nuclear stability	Some isotopes are stable; others decay (radioactivity)
Medical applications	Radioisotopes are used to diagnose and treat diseases, especially cancer
Dating ancient materials	Carbon-14 dating determines the age of archaeological artifacts
Energy production	Uranium-235 is used in nuclear power plants
Tracers in research	Radioisotopes track chemical reactions and biological processes
Potential harm	Radiation from radioisotopes can damage DNA and cause cancer

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Part 2: Explaining Isotopes – Same Protons, Different Neutrons

Atomic Structure Review

Subatomic Particle	Charge	Location	Role
Proton	Positive (+)	Nucleus	Determines the element (atomic number)

Subatomic Particle	Charge	Location	Role
Neutron	Neutral (0)	Nucleus	Determines the isotope (mass number)
Electron	Negative (-)	Orbitals around nucleus	Determines chemical behavior

How Isotopes Are Represented

Isotopes are written using the chemical symbol with the mass number (protons + neutrons) written as a superscript to the left.

Format: ^(Mass Number)Symbol

Isotope	Protons	Neutrons	Mass Number	Representation
Carbon-12	6	6	12	^{12}C
Carbon-13	6	7	13	^{13}C
Carbon-14	6	8	14	^{14}C

All three are carbon atoms (6 protons each). They differ only in the number of neutrons.

Examples of Isotopes for Common Elements

Element	Atomic No. (Protons)	Isotope	Neutrons	Mass Number	Stability
Hydrogen	1	Protium	0	1	Stable
Hydrogen	1	Deuterium	1	2	Stable
Hydrogen	1	Tritium	2	3	Radioactive
Uranium	92	Uranium-235	143	235	Radioactive

Element	Atomic No. (Protons)	Isotope	Neutrons	Mass Number	Stability
Uranium	92	Uranium-238	146	238	Radioactive
Chlorine	17	Chlorine-35	18	35	Stable
Chlorine	17	Chlorine-37	20	37	Stable

Key Points to Remember

Statement	Explanation
Same number of protons	Isotopes are the same element because proton count defines the element
Different number of neutrons	This is the only difference between isotopes of the same element
Same chemical properties	Chemical behavior depends on electrons, which are the same for all isotopes of an element
Different nuclear properties	Stability, radioactivity, and half-life vary between isotopes
Different mass numbers	Mass number = protons + neutrons; different neutrons mean different mass numbers

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Part 3: Natural and Human-Made Radioisotopes

Some isotopes are stable (they do not decay). Others are unstable and decay over time, releasing radiation. These are called **radioisotopes** (or radioactive isotopes). Radioisotopes can be found in nature or created artificially in laboratories and nuclear reactors.

Types of Radioisotopes

Type	How They Are Produced	Examples
Natural radioisotopes	Occur naturally in the environment; many are primordial (formed when the Earth was created)	Carbon-14 (^{14}C), Uranium-235 (^{235}U), Uranium-238 (^{238}U), Radium-226 (^{226}Ra), Radon-222 (^{222}Rn)
Human-made (artificial) radioisotopes	Produced in nuclear reactors, particle accelerators (cyclotrons), or as byproducts of nuclear reactions	Technetium-99m (^{99m}Tc), Cobalt-60 (^{60}Co), Iodine-131 (^{131}I), Cesium-137 (^{137}Cs)

Types of Radiation Emitted by Radioisotopes

When a radioisotope decays, it emits radiation. The three main types are alpha (α), beta (β), and gamma (γ) radiation.

Type of Radiation	Symbol	What Is It?	Charge	Penetrating Power	Shielding Material
Alpha (α)	α	Helium nucleus (2 protons + 2 neutrons)	+2	Very low (stopped by paper or skin)	Paper, clothing
Beta (β)	β	High-energy electron (or positron)	-1 or +1	Moderate (stopped by plastic or aluminum)	Plastic, aluminum
Gamma (γ)	γ	High-energy electromagnetic wave	0	Very high (stopped by lead or thick concrete)	Lead, concrete

Properties of Common Radioisotopes

Radioisotope	Type (Natural/Human-made)	Type of Radiation Emitted	Half-Life	Source / Use
Carbon-14 (^{14}C)	Natural	Beta (β)	5,730 years	Radiocarbon dating
Uranium-235 (^{235}U)	Natural	Alpha (α) and gamma (γ)	704 million years	Nuclear fuel, nuclear weapons
Uranium-238 (^{238}U)	Natural	Alpha (α)	4.5 billion years	Nuclear fuel (breeder reactors)
Radium-226 (^{226}Ra)	Natural	Alpha (α) and gamma (γ)	1,600 years	Formerly used in luminous paint
Radon-222 (^{222}Rn)	Natural (decay product of radium)	Alpha (α)	3.8 days	Health hazard in homes (soil gas)
Technetium-99m (^{99m}Tc)	Human-made	Gamma (γ)	6 hours	Medical imaging (most common)
Iodine-131 (^{131}I)	Human-made	Beta (β) and gamma (γ)	8 days	Thyroid cancer treatment and imaging
Cobalt-60 (^{60}Co)	Human-made	Gamma (γ)	5.27 years	Cancer radiation therapy (external beam)
Cesium-137 (^{137}Cs)	Human-made (nuclear fission product)	Beta (β) and gamma (γ)	30 years	Industrial radiography, medical irradiation

What Is Half-Life?

Half-life is the time required for half of the atoms in a sample of a radioisotope to decay. After one half-life, 50% of the original atoms remain. After two half-lives, 25% remain. After three half-lives, 12.5% remain, and so on.

Number of Half-Lives Elapsed	Fraction Remaining	Percentage Remaining
0	1/1	100%
1	1/2	50%
2	1/4	25%
3	1/8	12.5%
4	1/16	6.25%
5	1/32	3.125%

Examples of Half-Lives:

- Iodine-131: 8 days (short-lived; decays quickly)
- Carbon-14: 5,730 years (medium-lived; useful for dating)
- Uranium-238: 4.5 billion years (long-lived; primordial)

Part 4: Uses of Radioisotopes in Medicine

Radioisotopes have revolutionized medicine. They are used in two main ways: **diagnosis** (imaging to see what is happening inside the body) and **therapy** (treatment to destroy diseased tissue, especially cancer). Secondary sources such as medical journals, nuclear medicine textbooks, and government health agencies provide detailed information about these applications.

Diagnostic Uses (Imaging)

In diagnostic nuclear medicine, a small amount of radioisotope (called a tracer) is injected into the patient's body. The radioisotope accumulates in specific organs or tissues. A special camera (gamma camera or PET scanner) detects the radiation emitted and creates images.

Radioisotope	Procedure	What It Detects	Why This Isotope Is Used
Technetium-99m (^{99m}Tc)	Bone scan, heart scan, kidney scan	Bone cancer, heart disease, kidney function	Short half-life (6 hours); emits only gamma rays (lowest patient dose); attaches to many different molecules
Iodine-131 (^{131}I)	Thyroid scan	Thyroid disorders (hyperthyroidism, nodules)	Thyroid gland naturally concentrates iodine
Iodine-123 (^{123}I)	Thyroid scan	Thyroid function	Shorter half-life (13 hours); lower radiation dose than I-131
Fluorine-18 (^{18}F)	PET scan	Cancer, brain disorders, heart disease	Used in FDG (fluorodeoxyglucose) – cancer cells absorb more glucose
Thallium-201 (^{201}Tl)	Cardiac stress test	Blood flow to heart muscle	Behaves like potassium; accumulates in healthy heart muscle

Therapeutic Uses (Treatment)

In therapeutic nuclear medicine, larger doses of radioisotopes are used to destroy diseased tissue, particularly cancer cells. The radiation damages the DNA of the target cells, causing them to die.

Radioisotope	Treatment	How It Works	Why This Isotope Is Used
Iodine-131	Thyroid cancer and	Thyroid gland absorbs	Thyroid naturally

Radioisotope	Treatment	How It Works	Why This Isotope Is Used
(^{131}I)	hyperthyroidism	iodine; radioactive iodine destroys overactive thyroid cells or cancer cells	concentrates iodine; beta radiation kills nearby cells
Cobalt-60 (^{60}Co)	External beam radiation therapy (cancer treatment)	Gamma rays from Cobalt-60 are aimed at tumors from outside the body	Emits high-energy gamma rays; penetrates deeply to reach internal tumors
Strontium-89 (^{89}Sr)	Bone cancer (pain relief)	Strontium behaves like calcium and accumulates in bone	Beta radiation kills bone cancer cells and relieves pain
Yttrium-90 (^{90}Y)	Liver cancer, lymphoma	Delivered directly to tumor (selective internal radiation therapy)	Pure beta emitter (no gamma); treats tumor without exposing whole body
Radium-223 (^{223}Ra)	Prostate cancer that has spread to bone	Radium behaves like calcium and targets bone metastases	Alpha emitter (short range); kills cancer cells without damaging surrounding healthy tissue

Benefits vs. Risks of Medical Radioisotopes

Benefits	Risks
Non-invasive diagnosis (no surgery required)	Radiation exposure can increase cancer risk (small, but present)
Detects diseases early (often before symptoms appear)	Allergic reactions to tracers (rare)

Benefits	Risks
Provides functional information (not just anatomy)	Requires specialized equipment and trained personnel
Effective cancer treatment (targeted therapy)	Cost and availability of radioisotopes (some are expensive)
Can treat diseases that do not respond to surgery or drugs	Waste disposal (radioactive materials must be handled carefully)

How Scientists Use Secondary Sources to Learn About Medical Radioisotopes

Source	Information Provided
Medical journals (e.g., Journal of Nuclear Medicine)	Clinical studies on effectiveness of radioisotope treatments
Government health agencies (IAEA, FDA, Health Canada)	Safety guidelines, approved uses, regulations
Nuclear medicine textbooks	Detailed explanations of procedures and mechanisms
Hospital and clinic websites	Practical information for patients
Scientific databases (PubMed, Google Scholar)	Peer-reviewed research on new radioisotope applications

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Part 5: The Dual Nature of Radioisotopes – Beneficial and Harmful

Radioisotopes are powerful tools, but their power comes with risks. Understanding both the benefits and harms is essential for responsible use.

Beneficial Uses

Field	Application	Example
Medicine	Diagnosis and treatment of disease	Technetium-99m for bone scans; Iodine-131 for thyroid cancer
Energy	Nuclear power generation	Uranium-235 fission produces electricity
Archaeology	Dating ancient artifacts	Carbon-14 dating of fossils and historical objects
Agriculture	Improving crop yields; pest control	Gamma radiation kills pests; radioisotope tracers study fertilizer uptake
Industry	Detecting leaks; measuring thickness	Cesium-137 in thickness gauges; radioactive tracers in pipelines
Research	Tracing chemical reactions	Carbon-14 and tritium (^3H) used to study metabolic pathways

Harmful Effects

Harm	Explanation	Example
Cancer (radiation-induced)	Radiation damages DNA, potentially causing mutations that lead to cancer	Increased thyroid cancer after Chernobyl (due to Iodine-131)
Radiation sickness	High doses of radiation damage tissues and organs	Acute radiation syndrome from nuclear accidents
Environmental contamination	Radioisotopes can persist in the environment for thousands of years	Cesium-137 and Strontium-90 from nuclear accidents
Genetic mutations	Radiation can damage DNA in reproductive cells, affecting future generations	Increased birth defects in areas with high background radiation
Nuclear accidents	Uncontrolled release of radioisotopes	Chernobyl (1986), Fukushima (2011)

Harm	Explanation	Example
Nuclear weapons	Mass destruction and long-term environmental contamination	Atomic bombs on Hiroshima and Nagasaki (1945)

Safety Measures When Working with Radioisotopes

Safety Measure	Purpose
Time (limit exposure)	Less time exposed = lower dose received
Distance (increase distance)	Radiation intensity decreases with distance (inverse square law)
Shielding (use barriers)	Absorb radiation before it reaches the body (lead, concrete, plastic depending on radiation type)
Containment (seal sources)	Prevent radioisotopes from entering the environment
Monitoring (use dosimeters)	Measure radiation exposure to stay within safe limits

Key Terms

Term	Definition
Isotope	Atoms of the same element with the same number of protons but different number of neutrons
Radioisotope	An unstable isotope that decays, emitting radiation
Half-life	The time required for half of the atoms in a radioactive sample to decay
Alpha radiation (α)	Helium nucleus emitted from a radioactive nucleus; low penetration, high ionization
Beta radiation (β)	High-energy electron emitted from a radioactive nucleus; moderate penetration
Gamma radiation (γ)	High-energy electromagnetic wave emitted from a radioactive nucleus; high penetration
Tracer	A small amount of radioisotope used to track biological or chemical processes
PET scan	Positron emission tomography; medical imaging using radioisotopes
Irradiation	Exposure to radiation (medical use)
Contamination	The presence of unwanted radioactive material on surfaces or in the body

Review Questions

1. Explain what isotopes are. Your answer must include the number of protons, the number of neutrons, and why isotopes of the same element have identical chemical properties.

2. Complete the following table for the isotopes of carbon:

Isotope	Number of Protons	Number of Neutrons	Mass Number	Stable or Radioactive?
Carbon-12				
Carbon-13				
Carbon-14				

3. Using secondary sources (or the information provided in this lesson), identify two natural radioisotopes and two human-made radioisotopes. For each radioisotope, state the type of radiation it emits and its half-life.

4. Describe the three main types of radiation emitted by radioisotopes (alpha, beta, and gamma). For each type, state the composition, charge, penetrating power, and one example of a shielding material.

5. What is half-life? A sample of Iodine-131 (half-life = 8 days) has an initial mass of 100 grams. Calculate how many grams remain after 8 days, after 16 days, and after 24 days. Show your calculations.

6. Using secondary sources (or the information provided in this lesson), describe two different medical uses of radioisotopes: one diagnostic and one therapeutic. For each use, state the radioisotope used, the condition being diagnosed or treated, and why that particular radioisotope is suitable.

7. Explain why Technetium-99m is the most commonly used radioisotope in medical imaging. Your answer must include its half-life, type of radiation emitted, and why these properties are advantageous.

8. Radioisotopes can be both beneficial and harmful. Describe one benefit and one harm associated with the use of radioisotopes in medicine.

9. A hospital uses Cobalt-60 (half-life = 5.27 years) for cancer radiation therapy. Explain why this isotope is suitable for external beam therapy and what safety precautions must be taken when handling it.

10. Thinking question: Iodine-131 is used to treat thyroid cancer because the thyroid gland naturally concentrates iodine. However, Iodine-131 also emits gamma radiation that can expose other body tissues. Explain why the benefits of using Iodine-131 for thyroid cancer typically outweigh the risks.